- **A ripeness classification system based on key image features such as color, texture, and shape to determine the maturity level of tomatoes.**

No	Article Name	Year	Model / Method Used	Accuracy Achieved	Dataset Size	Ripeness Classes	Image Features Used	Future Work
1	Computational vision for tomato classification using a decision tree algorithm	2025	Decision Tree with CART (Gini index)	**83.6%**	60 tomatoes (20 per stage)	2 classes (Major vs. Low Purchasing Potential)	RGB, HSI, CIELab color models	Integrate with automated sorting & harvesting equipment; apply technique to other fruits
2	Tomato maturity recognition with convolutional transformers	2023	Hybrid CNN + Transformer (3-block encoder–decoder)	mAP **66.39%** (best across datasets)	≈2,700 images from multiple datasets	3 classes (Fully-ripe, Half-ripe, Un-ripe)	RGB with positional embeddings, multi-head attention	Add advanced loss functions, diversify datasets, include open-field images
3	Deep learning-based approach for tomato classification in complex scenes	2024	Deep Neural Network with pre-processing & maturity layers	Not specified	Internet images (multi-language collection)	5 classes (green, brittle, pink, pale-red, mature-red)	RGB with area of interest detection	Improve detection algorithms; integrate with automated harvesting systems
4	Tomato ripeness detection and fruit segmentation	2025	Improved YOLOv8s-seg with AOFRM, CMPRD,	mAP50 **91.3%** (boxes) / **91.2%** (masks)	1,061 high-resolution images	6 classes (Large/Small × fully-, semi-, immature)	Deformable convolution, multi-scale pooling	Validate with physicochemical tests; field trials; robot integration; model lightweighting

No	Article Name	Year	Model / Method Used	Accuracy Achieved	Dataset Size	Ripeness Classes	Image Features Used	Future Work
	(ACP-Tomato-Seg)		PSA modules					
5	Application of CNNs for Tomato Fruit Ripeness Identification	2025	Convolutional Neural Network	**92.67%**	1,000 images (Kaggle + field)	2 classes (ripe, unripe)	Image processing with data augmentation	Explore deeper CNNs, enlarge dataset, deploy in real time
6	AITP-YOLO: improved tomato ripeness detection model	2025	Enhanced YOLOv10s with 4-head detector, multi-scale fusion, Shape-IoU loss	mAP50 **92.6%**	Complex-background tomato dataset	Multiple ripeness stages	Multi-scale feature fusion; Shape-IoU bounding box optimization	Optimize for mobile/edge devices; enhance detection in cluttered scenes
7	Detection of Fruit Ripeness and Defectiveness Using Wide ResNet	2024	Wide ResNet (with XceptionNet & Inception-V4 comparison)	**97.87%**	2,589 beef-tomato images (5-fold CV)	6 classes (5 ripeness stages + defect)	Color, shape, texture based on USDA chart	Integrate with automated sorters; test on other fruits

No	Article Name	Year	Model / Method Used	Accuracy Achieved	Dataset Size	Ripeness Classes	Image Features Used	Future Work
8	Fruit ripeness identification using transformers	2023	Vision Transformer & Swin Transformer (with Mask R-CNN)	Precision **87.43%** (Swin Transformer)	Custom multi-fruit datasets	Multiple fruit types and ripeness levels	Transformer attention; bounding box detection	Refine transformer attention; build unified multi-fruit system
9	Tomato ripeness and stem recognition based on improved YOLOX	2025	YOLOX-SE-GIoU with SE focus module & GIoU loss	mAP **92.17%**	Tomato-fruit-and-stem dataset (class-imbalanced)	Multiple maturity levels + stem category	SE attention; GIoU loss optimization	Combine with robotic harvesters; fuse multi-sensor data

Model/method	Accuracy	Article	Year	Future Study
KNN Algorithm (**•KNN is an algorithm** **•It produces a model**)	Accuracy result of **91.25%** at the value of K = 7 with the 80 test data.	Classification Of Tomato Maturity Levels Based on RGB And HSV Colors Using KNN Algorithm	2022	The results of the accuracy obtained were quite good, but there were still classification errors because there were still tomato images that had a reflection of light with different intensity. This can cause errors in classification. **Further research to be able to pay attention to the quality of the image of tomatoes by minimizing the reflection of light when taking pictures**
K-Means Clustering method	*accuracy level for training data of 94% *accuracy level for test data of 92% Total accuracy level of both is **93%.**	Analysis of Tomato Ripeness by Color and Texture Using Cielab and K-Means Clustering	2023	

ANN model (Artificial Neural Networks)	To classify tomato ripeness into three distinct stages: **ripe, unripe, and semi-ripe**. The developed vision-based mobile robot achieved an overall classification accuracy of **94.5%,** with **ripe tomatoes** exhibiting the highest accuracy at **98%**, followed by **unripe (92%)** and **semi-ripe (89%).**	Development of a Vision-Based Mobile Robot with Artificial Neural Networks (ANN) for Classification of Tomato Ripeness	June 2025	Future work could focus on enhancing the system's adaptability to diverse environmental conditions, such as varying lighting or crop types, and optimizing energy consumption for prolonged field operations.
ML (machine learning)	The output of the ripeness regression models is divided into three classes based on their ripeness index and color magnitude to get the shelf-life of the tomatoes as Store,Sell, and Discount. The stacking technique is employed to achieve a final prediction of tomato shelf life with an impressive accuracy rate of **90.35%.**	Tomato ripeness and shelf-life prediction system using machine learning	February 2024	
Convolutional Neural Network (CNN) method.	Classified into three categories unripe, half-ripe, and ripe the model achieved a validation accuracy of **99.39%** and a loss value of 4.16%.	Classification of Tomato Ripeness Levels Using Convolutional Neural Network (CNN)	Jul. 16, 2025	In the future, the model can be further developed by incorporating optimization techniques or transfer learning to improve its efficiency and adaptability to data with more complex backgrounds or lighting conditions